PRESSURE CHECK FOR ALL LOAD COMBINATIONS

Project Name	:	G+2 RB FOOTING DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Footing-1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

Load Combinations (L / C):

L/C No	Load Definition
1	(LOAD 3: DL) +(LOAD 4: LL)
2	(LOAD 3: DL) +(LOAD 4: LL) +(LOAD 1: EQ X)
3	(LOAD 3: DL) +(LOAD 4: LL) -(LOAD 1: EQ X)
4	(LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)
5	(LOAD 3: DL) +(LOAD 4: LL) -(LOAD 2: EQ Z)

Footing No	=	FC1	
Column No	=	C1	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1450mm X 1250mm X 350mm	
Area of Footing (Af)	=	L X B	
	=	1.81	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.44	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.38	cum
Without considering Effect of Water Table			
Footing Self Wt., F1	=	15.86	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	15.86	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)

Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C1):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	186.3	0	0.05	111.69	111.38	0	200	0
2	116.43	2.71	24.19	143.25	2.73	25	250	0
3	256.16	-2.72	-24.08	220.08	80.09	25	250	0
4	139.83	30.49	2.18	161.27	10.52	25	250	0
5	232.76	-30.5	-2.07	212.28	62.06	25	250	0

Footing No	=	FC2	
Column No	=	C2	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1500mm X 1300mm X 350mm	
Area of Footing (Af)	=	L X B	
	=	1.95	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.49	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.42	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	17.06	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	17.06	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C2):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	258.67	1.27	3.74	152.86	129.94	0	200	0
2	254.96	1.35	33.33	221.15	57.85	25	250	0
3	262.37	1.19	-25.84	206.91	79.69	25	250	0
4	212.83	36.34	6.27	207.27	**28.51**	25	250	0
5	304.5	-33.8	1.21	**237.12**	92.7	25	250	0

Footing No	=	FC3	
Column No	=	C3	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1700mm X 1500mm X 425mm	
Area of Footing (Af)	=	L X B	
	=	2.55	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.72	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.64	cum
Without considering Effect of Water Table			
Footing Self Wt., F1	=	27.09	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	27.09	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C3):

L / C	Analysis Forces			Max. Soil	Min. Soil	SBC	Permissible	Loss of

	P (kN)	Mx (kNm)	My (kNm)	Pressures (kN/sqm)	Pressures (kN/sqm)	Increase (%)	Net SBC (kN/sqm)	Contact (%)
1	371.41	-0.72	1.19	159.14	153.41	0	200	0
2	445.26	-3.27	25.27	229.4	141.07	25	250	0
3	297.55	1.83	-22.89	165.75	88.88	25	250	0
4	310.36	41.76	3.64	195.83	**68.83**	25	250	0
5	432.46	-43.2	-1.26	**241.98**	118.45	25	250	0

Footing No	=	FC4	
Column No	=	C4	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1600mm X 1400mm X 375mm	
Area of Footing (Af)	=	L X B	
	=	2.24	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.6	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.52	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	21	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	21	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C4):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	305.15	6.84	0.83	158.65	132.56	0	200	0

2	233.16	9.95	27.34	182.43	**44.5**	25	250	0
3	377.14	3.73	-25.68	**233.11**	122.37	25	250	0
4	308.59	41.9	1.44	220.05	74.23	25	250	0
5	301.71	-28.22	0.22	191.74	96.39	25	250	0

Footing No	=	FC5	
Column No	=	C5	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1600mm X 1400mm X 375mm	
Area of Footing (Af)	=	L X B	
	=	2.24	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.6	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.52	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	21	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	21	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C5):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	297.31	7.05	4.38	162.28	121.93	0	200	0
2	294.48	7.14	36.77	223.15	58.53	25	250	0
3	300.14	6.96	-28.02	208.63	78.11	25	250	0
4	299.63	47.54	5.13	**232.54**	**53.74**	25	250	0

5	294.99	-33.43	3.62	203.97	78.17	25	250	0

Footing No	=	FC6	
Column No	=	C6	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1900mm X 1700mm X 450mm	
Area of Footing (Af)	=	L X B	
	=	3.23	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	1.02	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.92	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	36.34	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	36.34	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C6):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	560.93	7.79	-2.3	195.04	174.79	0	200	0
2	634.65	4.89	24.23	239	176.47	25	250	0
3	487.22	10.68	-28.83	204.04	**120.15**	25	250	0
4	574.96	56.25	-1.79	**246.21**	132.31	25	250	0
5	546.9	-40.68	-2.81	223.41	137.73	25	250	0

Footing No	=	FC7	
Column No	=	C7	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1550mm X 1350mm X 375mm	
Area of Footing (Af)	=	L X B	
	=	2.09	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.54	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.47	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	19.62	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	19.62	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C7):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	285.83	7.69	0.71	161.69	130.26	0	200	0
2	207.4	10.79	28.61	189.22	27.76	25	250	0
3	364.26	4.58	-27.2	249.68	117.22	25	250	0
4	272.7	43.08	0.24	219.89	59.5	25	250	0
5	298.96	-27.71	1.17	206	98.49	25	250	0

Footing No	=	FC8
Column No	=	C8
Design Code	=	IS 456 : 2000 + IS 13920 : 2016

Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1700mm X 1400mm X 375mm	
Area of Footing (Af)	=	L X B	
	=	2.38	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.67	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.56	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	22.31	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	22.31	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C8):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	292.59	10.92	5.14	157.76	106.86	0	200	0
2	292.35	11.05	43.61	227.12	37.3	25	250	0
3	292.83	10.8	-33.34	208.46	56.37	25	250	0
4	273.72	70.18	4.44	**236.46**	**12.3**	25	250	0
5	311.46	-48.33	5.83	222.42	58.07	25	250	0

Footing No	=	FC9	
Column No	=	C9	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm

Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1800mm X 1600mm X 425mm	
Area of Footing (Af)	=	L X B	
	=	2.88	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.86	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.77	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	30.6	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	30.6	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C9):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	447.29	9.54	-1.22	178.56	153.3	0	200	0
2	527.58	6.7	26.69	**236.32**	151.3	25	250	0
3	366.99	12.38	-29.13	190.31	**85.79**	25	250	0
4	423.47	58.42	-1.42	227.13	88.19	25	250	0
5	471.1	-39.34	-1.01	221.05	127.35	25	250	0

Footing No	=	FC10	
Column No	=	C10	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%

Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1700mm X 1500mm X 425mm	
Area of Footing (Af)	=	L X B	
	=	2.55	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	0.72	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.64	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	27.09	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	27.09	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C10):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	366.79	2.56	-0.61	158.96	149.96	0	200	0
2	278.16	6	29.35	174.04	65.37	25	250	0
3	455.41	-0.88	-30.57	238.39	140.05	25	250	0
4	423.77	33.29	-2.65	227.04	126.58	25	250	0
5	309.8	-28.18	1.42	173.35	90.88	25	250	0

Footing No	=	FC11	
Column No	=	C11	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1950mm X 1650mm X 425mm	

Area of Footing (Af)	=	L X B	
	=	3.22	sqm
Section Modulus, Zxx	=	B x L^2 / 6	
	=	1.05	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.88	cum

Without considering Effect of Water Table

Footing Self Wt., F1	=	34.19	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	34.19	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C11):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	500.41	0.79	5.03	172.6	159.7	0	200	0
2	503.41	1.19	45.53	219.68	114.49	25	250	0
3	497.41	0.4	-35.46	205.67	124.76	25	250	0
4	561.21	54.6	2.46	**240.05**	130.05	25	250	0
5	439.6	-53.01	7.61	206.55	**87.96**	25	250	0

Footing No	=	FC12	
Column No	=	C12	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Depth of founding layer	=	1.5	m
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Design cross section by	=	Average pressure	
Footing Type	:	Pad	
Footing Size (L X B X D)	:	1900mm X 1700mm X 450mm	
Area of Footing (Af)	=	L X B	
	=	3.23	sqm
Section Modulus, Zxx	=	B x L^2 / 6	

	=	1.02	cum
Section Modulus, Zyy	=	L x B^2 / 6	
	=	0.92	cum
Without considering Effect of Water Table			
Footing Self Wt., F1	=	36.34	kN
Soil Weight, S1	=	0	kN
Total Weight, W1	=	F1 + S1	
	=	36.34	kN

Note:
P1 (for Max Pressure) = P + W1
P2 (for Min Pressure) = P + W2
Max Soil Pressure = P1/A + Abs(Mx/Zxx) + Abs(My/Zyy)
Min Soil Pressure = P2/A - Abs(Mx/Zxx) - Abs(My/Zyy)
Note1: For Eccentric Footings, Mx will be replaced by Mx + Mex & My + Mey.
Note2 : For Footings of Irregular Shaped Columns, Mx will be replaced by Mx + P x Exo & Mx + P x Eyo.

Soil Pressure Checks Table (C12):

L / C	Analysis Forces			Max. Soil Pressures (kN/sqm)	Min. Soil Pressures (kN/sqm)	SBC Increase (%)	Permissible Net SBC (kN/sqm)	Loss of Contact (%)
	P (kN)	Mx (kNm)	My (kNm)					
1	538.91	1.35	0.11	179.53	176.66	0	200	0
2	623.72	-2.49	30.61	240.24	168.46	25	250	0
3	454.1	5.19	-30.39	190.12	113.56	25	250	0
4	610.51	45.28	-2.62	**247.39**	153.13	25	250	0
5	467.31	-42.58	2.84	200.67	**111.19**	25	250	0